of sex determination, the alphabets 'Z' and 'W' are used here instead of X and Y respectively. Heterogametic females are of two types, ZO-ZZ type (eg. Moths, butterflies and domestic chickens) and ZW-ZZ type (eg. Gypsy moth, fishes, reptiles and birds).

Sex determination in human beings

Genes determining sex in human beings are located on two sex chromosomes, called allosomes. In mammals, sex determination is associated with chromosomal differences between the two sexes, typically XX females and XY males. 23 pairs of human chromosomes include 22 pairs of autosomes (44A) and one pair of sex chromosomes (XX or XY). Females are homogametic producing only one type of gamete (egg), each containing one X chromosome while the males are heterogametic producing two types of sperms with X and Y chromosomes. An independently evolved XX: XY system of sex chromosomes also exist in *Drosophila* **(Fig. 4.2)**.

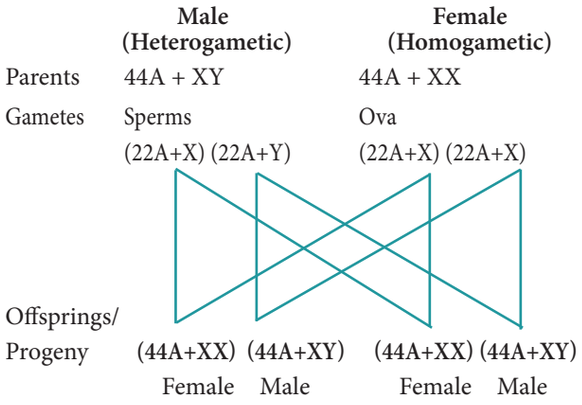


Fig. 4.2 Sex determination in human beings

The Y Chromosome and Male Development

Current analysis of Y chromosomes has revealed numerous genes and regions with potential genetic function; some genes with or without homologous counterparts are seen on the X. Present at both ends of the Y chromosome are the pseudoautosomal regions (PARs) that are similar with regions on the X chromosome which synapse and recombine during meiosis. The remaining 95% of the Y chromosome is referred as the Non - combining Region of the Y (NRY). The NRY is divided equally into functional genes (euchromatic) and non functional genes (heterochromatic). Within the euchromatin regions, is a gene called **Sex determining region Y (SRY)**. In humans, absence of Y chromosome inevitably leads to female development and this SRY gene is absent in X chromosome. The gene product of SRY is the testes determining factor (TDF) present in the adult male testis.

4.4.1 Genic balance in Drosophila

Genic balance mechanisms of sex determination in *Drosophila* was first studied by C.B. Bridges. In *Drosophila*, the presence of Y chromosome is essential for the fertility of male sex, but does not determine the male sex. The gene for femaleness is located on the X chromosome and those for maleness are located on the autosomes. When geneticist C.B. Bridges, working with *Drosophila*, crossed a triploid (3n) female with a normal male, he observed many combinations of autosomes and sex chromosomes in the offspring. From his results Bridges in 1921 suggested that sex in *Drosophila* is determined by the balance between the genes for femaleness located on the 'X' chromosomes and those for maleness located on the autosomes. Hence the sex of an individual is determined by the ratio of its X chromosome to that of its autosome sets. This ratio is termed **sex index** and is expressed as:

$$\text{Sex index} = \frac{\text{Number of X Chromosomes}}{\text{Number of Sets of Autosomes}} \left(\frac{X}{A}\right)$$

Change in this ratio leads to a changed sex phenotype. The results obtained from a cross between triploid female *Drosophila* (3A:3X) with a diploid male (2A: XY) is shown in **tables 4.2. and 4.3.**

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